

BREAKHOOK · CALCULATION

BreakHook sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Fork prop at mid-length added under Amish's decision 6A (BHK-DDR-003); droop, reach, prop load, buckling, prop mass, insulated path to the prop holder [F13 to F24, H3]; kit cost re-run
0.1	2026-10-03	Amish Chadha	First issue for TRL 3 on the constructable design (pull needed, hauling geometry, hook plate, welds, rope system, pole placement, joints, insulation, deployment time, mass and cost)

BreakHook sizing calculations

The constructable design meets its strength, mass, insulation and reach targets on paper: the hook plate keeps a factor of 2.3 or more on yield at the 6 kN proof load, the pole and hook head weigh 7.82 kg against 8 kg, and haulers stand 10.8 m from the wall. With the fork prop at mid-length (BHK-DDR-003) the hook droops about 0.27 m instead of 0.66 m, and nobody lifts the pole: the prop carries 100 N into the ground. Two results still need care: the prop holder stands about 2.9 m from the wall while the hook is set, nearer than the 4 m of R1, and deployment comes out at exactly the 3 minute target. Times and pulls are estimates that a TRL 4 trial on a purpose-built test frame must confirm.

Safety: This note sizes a tool that pulls down a structure with a rope. The numbers assume the dwelling has been checked empty, nobody is inside the fall zone, the pole has been withdrawn before the haul and there are no overhead lines within 3 m. Nothing in this note has been built or tested. Proof loads are applied to each hook head before issue, never to a person or an occupied structure.

■ Scope and method

First-order hand calculations, run by `docs/04-calcs/sizing.py`, which takes every dimension and mass from the model (`cad/src/model.py`) and writes `docs/04-calcs/results.csv`. Each result has a reference in brackets, for example [A1]. Loads are static; no dynamic or impact factor is applied beyond the 2 x proof load, which is stated where it matters.

■ Assumptions

Table 1. Assumptions

#	Assumption	Value	Basis
1	Target dwelling	Single storey, 3.0 x 3.0 m, walls 2.4 m, light timber frame clad in corrugated steel sheet, 350 kg	Typical informal dwelling (BHK-PRB-001); to be confirmed with the co-design partner
2	Racking resistance of a nailed, clad, unbraced frame	0.5 kN per metre of side wall	Engineering judgement for light cladding; to be measured on the TRL 4 test frame
3	Sustained pull per hauler on a rope with toggles	300 N	Conservative for adults on dry ground; peak pulls are higher
4	Pull rope breaking strength	25 kN or more, spliced eye 90 %	Bought to this specification (BOM 13)
5	Leader breaking strength	20 kN or more, pressed ferrules 90 %	6 mm 6x19 galvanised wire rope (BOM 12)
6	Hook plate	S355, yield 355 MPa, 10 mm	BOM 1
7	Fibreglass pole tube	E = 20 GPa along the tube; flexural strength 200 MPa; density 1,900 kg/m ³ ; foam 60 kg/m ³	Low end of published values for pultruded round tube; to be confirmed from the maker's data sheet
8	Placement geometry	Hook at 3.0 m, front hand 1.1 m high and 4.0 m from the wall	R1
9	Store distance	Kit stored within 100 m of the dwellings it covers; walking 1 m/s with the kit	Community fire plan rule
10	Nylon 66 shear strength	40 MPa	Conservative for a moulded pin
11	Epoxy allowable lap shear	5 MPa	One third of a typical 15 MPa structural epoxy
12	Fork prop	Holds the pole at mid-length, leaning about 13° with its top toward the wall and its foot on the ground; the rear hand holds the butt 100 mm from its end; both are simple supports	BHK-DDR-003

■ A. How hard to pull (R2, R7, R13)

Pulled at the top of the near wall, a rigid 350 kg dwelling tips over its near edge at 2.15 kN [A1]. A clad frame more often racks: at 0.5 kN per metre of the two 3 m side walls it needs about 3.0 kN [A2]. The working pull is set at 3 kN [A3] and every pulling part is sized for a 6 kN proof load [A4]. At 300 N each, ten haulers give the working pull [A5]; eight are enough for the overturning case [A6].

■ B. Hauling geometry (R3)

The first toggle is 11 m along the line from the hook head. With the hook at 3.0 m and hands at 1.0 m, the nearest hauler stands 10.8 m from the wall [B1], against 4.5 m required by R3 [B2]. The rope rises at 10.5° [B3], so 98.3 % of the pull acts along the ground [B4]. The ten toggles run to 20 m along the 26.5 m line [B5], leaving a 6.5 m tail [B6].

■ C. Hook plate at the 6 kN proof load (R2, R11)

Table 2. Hook plate stresses at the proof load

Ref	Check	Result	Factor on yield (355 MPa)
C2	Arm root bending, 75 mm deep beam (lever 58 mm)	97.9 MPa	3.6 at proof, 7.3 at working
C3	Arm root bending, 100 mm deep beam (lever 90 mm)	153 MPa	2.3 at proof, 4.6 at working
C4	Shank tension and bending (rope line 0.5 mm off the arm load line)	18.1 MPa	19.7
C5	Shackle hole tear-out, two 23.5 mm planes	12.8 MPa	large
C6	Shackle pin bearing on the plate	54.5 MPa	6.5
C7	Rope tab root bending	37 MPa	9.6

The arm is raked back 16.7° [C1] so a pull draws the beam deeper into the hook. The shackle hole is placed so the rope line passes through the middle of a 75 mm beam; the shank then carries almost pure tension [C4]. The throat takes beams up to 150 mm wide at the shank and 128 mm wide 100 mm down [C8, C9], and up to 117 mm deep [C10].

■ D. Socket welds

The rope pulls the plate, never the socket [D2]. The welds carry only placement loads: a 200 N side push at the spike tip gives 4.6 MPa in the four 6 mm fillet welds [D1].

■ E. Rope, leader and shackles (R2)

Factors at the 3 kN working pull: pull rope 7.5 [E1], wire rope leader 6.0 [E2], shackle working load limit 3.3 times the pull [E3]. Each prusik cord carries about 300 N [E4] against a breaking strength near 7 kN. The proof load of 6 kN is within the shackle's working load limit.

■ F. Pole placement (R1, R5, R8)

Setting the hook at 3.0 m from 4.0 m away puts the pole at 25.4° [F1]. The front hand is 1.27 m from the butt [F2], below the hand band at 1.55 m [F3]. The pole, hook head, leader and the rope hanging from it weigh 84 N [F4]. Held with the hands 1.17 m apart, the rear hand pushes down with 168 N [F5] and the front hand lifts 252 N [F6], so the pole is handled by a team of two: one lifts, one holds the butt down. The tube stress at the front hand is 50.6 MPa, a factor of 4.0 on its flexural strength [F7].

The tube is flexible: with $E = 20$ GPa the hook droops about 658 mm below the straight line [F8] (second moment of area 8.91 cm^4 [F9]). Held out by hand, the team would have to aim about 0.7 m high; a stiffer tube would add mass against R5. The fork prop below takes most of this droop out.

With the fork prop (BHK-DDR-003)

Amish's decision 6A adds a light fork prop that holds the pole at its mid-length, 2,944 mm from the butt [F13]. In the R1 placement the pole's underside is 1.79 m above the ground there [F14]; the prop is set at its third hole (fork notch 1.84 m above the foot) and leans 12.8° with its top toward the wall [F15]. The pole then rests on two supports, the prop and the rear hand at the butt, with the hook end overhanging the prop.

- **Droop:** the hook droops 274 mm below the line through the two supports [F16], against 658 mm when the pole is held out by hand [F8]. The team aims about 0.3 m high and lowers the hook onto the beam.
- **Effort:** the prop carries 100 N into the ground through its foot [F17]; the rear hand pushes down with only 24 N [F18], against 252 N lifted and 168 N pushed down by hand [F5, F6]. The prop holder only steadies the prop.
- **Reach:** the hook is set at 3.0 m. The rear hand is 5.06 m from the wall [F20]. The prop foot is 2.89 m from the wall [F19], and the prop holder stands at or just behind it, so the nearest person is about 2.9 m from the wall while the hook is set, against 4.0 m for the front hand in R1.
- **Strength:** the prop's upper tube alone buckles at 822 N over the whole prop length, 8 times the prop load [F21].
- **Mass:** the prop weighs 1.08 kg [F22] and is carried separately, so the pole and hook head stay at 7.82 kg (R5). Closed, the prop is 1.24 m long (R8).

Moving the prop toward the butt until its foot is 4.0 m from the wall puts it 1,577 mm from the butt [F23]; the hook then droops 720 mm [F24], more than by hand, because nearly all the pole overhangs the prop. The prop only helps near mid-length.

Mass: the pole and hook head weigh 7.82 kg [F10], 0.18 kg under the 8 kg limit [F11]. The longest piece to carry is a section with its sleeve, 1.95 m [F12].

■ G. Pole joints and release (R12)

The friction ring is wrapped until a 50 N pull frees the pole from the hook head [G1]. That is about four times the hook head's own weight along the pole at 25° [G2] and needs only 0.018 MPa of contact pressure [G3]. A nylon pin in double shear holds 6.3 kN [G4], about 126 times the release force [G5]; the pins never carry the rope pull. The bonded sleeve holds about 105 kN [G6].

■ H. Insulation (R4, R14)

There is no metal between the mouth of the socket and the top of the hand band: 3.67 m of foam-filled fibreglass with nylon pins [H1], against 3.0 m required [H2]. The prop holder's hand is 3.27 m of fibreglass, HDPE and nylon from the socket mouth: along the pole to the fork, then down the prop to the top of its red hand band [H3]. Whether that length insulates depends on the bought tube's certificate (R4). The hook, leader and shackles are metal and a wet rope conducts.

Safety: The insulated length is a second line of defence only. The rule on the check card stands: never raise the pole within 3 m of any overhead line, whatever the pole is made of.

■ I. Deployment and pull time (R6, R7)

Table 3. Deployment estimate (pole team of two)

Step	Seconds
Cut seal, unlock, lift out one set	20
Carry to the fire edge, 100 m at 1 m/s	100
Join three sections, two pins	20
Hook head onto the pole	5
Set the hook on the beam	25
Withdraw the pole, step out of the fall zone	10
Total	180 (3.0 min) [I1]

The other two of the four lay out the rope and run the check card at the same time. The pole team is the rear handler and the prop holder: the prop is stored at the hole marked for the community's usual wall height and goes under the pole in place of the front-hand lift, so the estimate is unchanged; a timed drill at TRL 4 must confirm it. From hook set to frame down: crew on the toggles and the caller's zone check 30 s, slack taken up 10 s, pull 30 s, about 1.2 minutes [I2].

■ L. Results against every requirement

Table 4. Requirements against the calculations (requirement text in BHK-REQ-001)

ID	Target	Result	Status
R1	Hook at 3 m from 4 m away	25.4° pole on the fork prop: droop 0.27 m (0.66 m by hand), rear hand 24 N; rear hand 5.06 m and prop holder 2.9 m from the wall [F13 to F20]	Hook placement met on paper with much less droop; the prop holder stands 1.1 m nearer than the 4 m front-hand distance
R2	3 kN working, 6 kN proof	Plate factor 2.3 on yield at proof; rope 7.5, leader 6.0, shackle 3.3 [C, E]	Met
R3	Haulers at 1.5 x height or more	10.8 m against 4.5 m [B1]	Met
R4	Non-conductive handled length, test voltage set	Foam-filled tube with a maker's ASTM F711 certificate [H1]	Met by specification; confirm the certificate when bought
R5	8 kg or less	7.82 kg [F10]; the 1.08 kg prop is carried separately [F22]	Met, 0.18 kg margin
R6	3 min or less with four people	3.0 min [I1]	Met on estimate, no margin; timed drill at TRL 4
R7	Under 2 min to bring down a test frame	About 1.2 min [I2]	Estimate only; timed trial at TRL 4
R8	No piece over 2 m	1.95 m [F12]; prop 1.24 m closed	Met
R9	12 months outdoor storage	Material choices only	Not verifiable at TRL 3
R10	Kit cost against the USD 2,000 target	USD 1,170 with two fork props (bom/bom.csv)	USD 830 under the value-engineering target
R11	Beams up to 100 x 120 mm	117 mm deep, 128 to 150 mm wide [C8 to C10]	Met
R12	Pole frees from the head before the haul	50 N release [G1]	Met by design; set by trial
R13	300 N or less per hauler	300 N with ten haulers [A5]	Met at the limit
R14	3.0 m or more insulated length	3.67 m [H1]; 3.27 m to the prop holder's hand [H3]	Met

■ Checks against the TRL 1 figures

The TRL 1 precis gave a 4 to 6 m pole; the design is 5.9 m overall. The 8 kg mass target holds with 0.18 kg to spare only because the hook plate was trimmed to a 36 mm shank (BHK-DDR-002). The 3 kN working pull stands; the calculation shows it needs about ten people, not four, so the four-person team of R6 deploys the kit and the hauling crew comes from residents on the scene.